VARUNA AquaHalo — Master Technical & Presentation Document

Vishwakarma Award Hackathon — Team ID: [INSERT TEAM ID] · Track: [INSERT OFFICIAL TRACK]

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1. Executive Summary

VARUNA AquaHalo is a small, remotely-controlled, twin-float catamaran that collects floating surface debris (bottles, wrappers, cups, thermocol, caps) from calm ponds, lakes and contained water bodies using a **conveyor-free, passive collection mechanism** — a V-shaped intake, a static retention baffle, and a removable mesh basket — instead of the powered conveyor belts used by almost every comparable Indian student project.

Design philosophy: float → stabilize → retain debris reliably, before any smart feature. Control is phone-based, using the team's already-owned Arduino Uno (motor logic) and ESP32-CAM (camera + WiFi command link). New-purchase budget target: **₹8,000–₹10,000**, with a floor version achievable near ₹5,000.

What this is NOT: an autonomous ocean-cleaning robot, a water-purification device, a microplastic remover, or an AI waste-recognition system. All of these are explicitly excluded from Version 1 and this document says so wherever relevant.

Status key used throughout: `[VERIFIED]` = confirmed by research/search · `[ASSUMPTION]` = engineering estimate, needs build/test to confirm · `[TARGET]` = design goal, not yet achieved · `[MEASURED]` = real test result · `[FUTURE]` = post-V1 only.

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2. Problem Statement & Objectives

Problem: Floating plastic and solid debris accumulates on pond, lake and canal surfaces. It is currently removed by hand (net from the bank or a rowboat) by pond-maintenance staff, campus grounds teams, or environmental-cleanup volunteers — slow, physically repetitive, and limited to what is reachable from shore or a crewed boat.

Target beneficiaries (presented as target users, not confirmed customers): pond/lake maintenance teams, campus facilities staff, local environmental-cleanup groups, organizations running supervised surface-cleaning drives.

Why manual collection is inconvenient: debris spreads across the whole surface, not just the shoreline; reaching the centre of a pond requires wading or a boat; collection is repetitive and time-consuming per unit of debris gathered `[baseline values: To be measured during field testing — no invented labour/time figures]`.

Why a complex mechanism is wrong for this project: a powered conveyor belt (used in nearly every comparable Indian student project — see Section 8) adds motor count, cost, and jam risk without, in the literature reviewed, any reported measurement of actual collection efficiency. A passive, conveyor-free mechanism removes that failure mode at lower cost.

Precise scope of V1:

- Targets floating macro-debris only (bottles, wrappers, cups, thermocol, caps)
- Operates in a contained, supervised test pond/tank
- Manually, remotely controlled — not autonomous
- Demonstrates measured collection efficiency and retention-during-turning, not just "it floats and picks something up"

Explicitly excluded claims: AquaHalo does not clean an entire lake autonomously, does not remove submerged waste, does not remove microplastics, and does not purify water.

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3. Slide-by-Slide Content (Slides 1–17)

Each slide entry below gives: Title · Main message · Content · Visual needed · Status/validation note · Speaker-note gist. Full image-generation prompts for the CAD/diagram visuals are in Section 6 and Section 4 (these are reused across slides rather than repeated per-slide to avoid redundancy, per the "do not repeat explanations" instruction).

Slide 1 — Project Overview

Message: A low-cost, conveyor-free floating waste collector built around a passive retention mechanism, not an added sensor. **Content:** Project title (VARUNA AquaHalo), Team ID / Track placeholders, one-line problem summary, core innovation statement, isometric concept render. **Core innovation statement (defensible, from Section 7 prior-art review):**

> "A conveyor-free floating waste collector using a modular V-intake and a passive, no-moving-parts retention baffle, adapted from a published macro-scale 'Tesla-valve-inspired' debris-retention concept and miniaturized to a sub-₹10,000, student-buildable scale — with intake and retention performance evaluated experimentally rather than assumed." **Visual:** Isometric render (see Section 6 CAD prompt). **Status:** `[ASSUMPTION — concept render, not yet fabricated]`. **Speaker note:** "We're not claiming to have invented floating waste collection — we're claiming a specific, testable improvement over the conveyor-belt pattern every similar student project uses."

Slide 2 — Problem Statement

Message: Manual debris collection on small water bodies is slow and effort-heavy; existing student solutions solve it with a jam-prone conveyor. **Content:** operational problem, beneficiary segment, why manual collection is inconvenient, why a complex mechanism is the wrong call at this budget/timeline, precise scope of AquaHalo (as in Section 2 above). **Visual:** Simple problem→solution diagram (manual collection icon → AquaHalo icon, three arrows labelled "slow," "shore-limited," "inconsistent"). **Quantified claims:** none invented; baseline values marked `[To be measured during field testing]`.

Slide 3 — Before State: Current User Journey

Message: Manual collection is a five-step repetitive loop with real friction points. **Content (storyboard):** 1) Operator inspects surface → 2) manually collects floating debris → 3) transfers to container → 4) repeats → 5) sorts and disposes. **Pain points:** repeated physical effort, time gathering scattered debris, difficulty reaching some areas, manual handling of wet waste, inconsistent collection performance. **Visual:** 5-panel storyboard, simple line-art figures. **Note:** No time/labour numbers invented; plan to baseline-measure during testing if pursued.

Slide 4 — After State: AquaHalo User Journey

Message: The same job, done from the shore, with measurable performance. **Content (storyboard):** 1) Operator inspects test area → 2) robot checked and placed on water → 3) phone control connected → 4) twin motors move the boat → 5) V-guides direct debris into basket → 6) retention mechanism reduces escape → 7) robot returned → 8) basket

removed, emptied, cleaned → 9) test performance recorded. **Visual:** 9-panel storyboard + Before/After split comparison. **Status:** No percentage improvement claimed before testing — target metrics only, marked `[Pending prototype validation]`.

Slide 5 — System Block Diagram

Message: A deliberately simple architecture — one board for vision/comms, one for motor logic, no conveyor, no unapproved sensors. **Content:** Mechanical block (floats, frame, V-intake, baffle insert, retention mechanism, basket) · Actuation/power block (battery, fuse, cutoff, regulation, driver, 2 motors) · Electronics block (Arduino Uno, ESP32-CAM, phone, camera feed, defined serial link) · Firmware block (safe power-on state, command receive/validate, drive logic, comms timeout, e-stop, operator recovery). **Visual:** full block diagram — see Section 4 ASCII reference and Section 6 image-prompt. No sensor beyond the camera is drawn (explicitly excludes turbidity/TDS/temperature/GPS per Section 2.4 exclusions). **Status:** `[ASSUMPTION — architecture validated on paper/breadboard, not yet on full hardware]`.

Slide 6 — Target Specifications

Parameter	Preliminary target	Status	Verification method
Overall dimensions	600 × 450 mm	`[ASSUMPTION]`, subject to CAD check	CAD model + physical measurement
Dry mass	~2.4 kg	`[ASSUMPTION]` (Section 5 BOM-based estimate)	Weigh built prototype
Safe payload (waste)	~1.0–1.5 kg	`[TARGET]`	Loaded float test
Float displacement (both floats, full submersion)	17.1 L	`[VERIFIED — geometric calc]`	—
Draft at operating mass (~3.5–3.9 kg)	~37 mm	`[ASSUMPTION — calculated]`	Loaded flotation test
Freeboard at operating mass	~143 mm	`[ASSUMPTION — calculated]`	Loaded flotation test

Parameter	Preliminary target	Status	Verification method
Propulsion	2× reversible rear motors (propeller, paddle-wheel alternative considered)	`[TARGET]`	Bench thrust test
Battery	2S Li-ion, ~7.4V, ~2200mAh	`[TARGET]`	—
Estimated runtime	Theoretical only — depends on measured current draw	`[TARGET – not yet calculable without hardware]`	Timed discharge test
Collection efficiency	To be measured	`[Testing pending]`	(items recovered / items released) × 100
Retention after turning	To be measured	`[Testing pending]`	(items retained after 360° turn / items collected before turn) × 100
Control range	To be measured (ESP32-CAM WiFi, open-area)	`[Testing pending]`	Range walk test
Total new-purchase budget	₹8,000–₹10,000 target (floor ~₹5,000)	`[TARGET]`	Section 7 BOM

No unrelated benchmark figures (e.g. 25 kg payload, 1.5 m/s, 4-hour runtime) are used — none are supported by this hardware or budget.

Slide 7 — 3D CAD and Assembly

Message: A fixed, consistent geometry — twin floats, twin rear motors, V-intake, retention baffle, rectangular basket — carried through every view and every image prompt. **Content:** isometric assembled view, top view, side view with waterline marked, exploded assembly view, labelled callouts, initial dimensions, assembly order (see Section 10 roadmap for sequence). **Required consistent elements:** exactly 2 floats; exactly 2 rear

propulsion motors; front V-intake; retention baffle (no hinge; see Section 7 novelty note on why the flap was replaced); long rectangular mesh basket; electronics enclosure above waterline; removable basket hatch. No conveyor, no GPS, no extra motors. **Label requirement:** "Concept CAD — subject to prototype validation" on every render until native CAD exists. **Full image-generation prompt:** see Section 6.

Slide 8 — Materials, Loads and Structural Integrity

Message: Material choices are driven by weight, water resistance, repairability and cost — not aesthetics.

Component	Options compared	Recommendation	Why
Floats	Sealed PVC pipe / HDPE pipe / sealed containers / foam-filled	**Sealed PVC pipe, end-capped**	Matches target float geometry, solvent-weldable, leak-testable, widely available in India
Frame	Plywood / acrylic sheet	**Plywood or 3–5mm acrylic**	Low cost, easy to drill/mount, acceptable water tolerance with sealant
Intake baffle	PETG vs acrylic	**PETG for first build**	More impact-tolerant than acrylic; acrylic is more brittle under repeated handling during testing iterations
Basket mesh	Stainless kitchen mesh	3–5 mm gap, size TBD by target debris	Must be re-validated against actual debris set — fine mesh clogs with leaves
Fasteners	Standard SS/plated bolts, nylon fasteners near electronics	Mixed	Avoid bare steel near the waterline long-term

Preliminary calculations (full detail in Section 6): float displacement, draft, freeboard, basket load support, frame loading, propulsion mount loading, centre-of-mass with uneven basket loading.

FEA status: not performed. No fabricated stress contours are shown. Instead: critical load-bearing areas identified (basket support rails, float-to-frame mounting points, motor mounts), the CAD/FEA workflow needed (static load case on basket rails at 1.5 kg eccentric load; boundary conditions = frame mount points fixed; mesh-convergence check if FEA is eventually run), and prototype tests required to validate (physical loaded-basket deflection test substitutes for FEA in V1).

Slide 9 — Circuit and Control Architecture

Message: ESP32-CAM handles vision/WiFi only; Arduino Uno handles motor logic — kept separate because the ESP32-CAM has very limited free GPIO once the camera and WiFi stack are running. **Content:** battery → fuse → physical cutoff → regulation → motor driver (L298N or equivalent) → 2 motors; Arduino Uno (digital pins → driver logic inputs) ↔ serial link ↔ ESP32-CAM (WiFi → phone webpage + camera stream); shared ground between boards. **Explained explicitly:** motor power rail is separate from logic power rail; driver must be rated for each motor's *stall* current, not just running current `[per-motor stall current: to be measured on the actual selected motor before final driver selection]`; motors are never wired directly to microcontroller pins; ESP32-CAM power draw checked against the shared supply before finalizing regulation. **Pin assignments:** withheld until the actual motor driver model and Arduino/ESP32-CAM board variant are confirmed — `[INSERT PIN MAP once hardware selected]`. **Visual:** full schematic — see Section 6.

Slide 10 — Itemized BOM and Procurement

See Section 7 for the full table (reused here, not duplicated) — target new-purchase total within ₹8,000–₹10,000, floor version near ₹5,000. All prices are supplier-listed estimates requiring confirmation before purchase, not guaranteed quotes.

Slide 11 — Maker Journey: Design and Fabrication

Message: A defined ten-step build sequence from requirements to water trials — nothing skipped, nothing fabricated as "done" before it is. **Sequence:** 1) requirements definition → 2) CAD modelling → 3) buoyancy/load calculations → 4) mechanical fabrication → 5) intake and basket assembly → 6) electronics mounting → 7) wiring and firmware → 8) dry bench tests → 9) float and leak tests → 10) controlled water trials. **Evidence checklist to actually capture (not assumed to exist yet):** CAD application + feature tree, native CAD files,

assembly views, wiring schematic, firmware repository, bench-test photographs, prototype photographs, test logs. `[INSERT once each item exists — none fabricated here]`.

Slide 12 — Software and Repository Evidence

Message: Real files, real repository, real timestamps — nothing staged. **Required files:**

- *Mechanical:* native CAD source, STEP export, assembly/part files
- *Electronics:* native schematic, PCB files only if a PCB is actually designed, BOM
- *Firmware:* Arduino source, ESP32-CAM source, README, pin map, control-protocol doc, test logs
- *Simulation:* calculation sheets; FEA files only if FEA is actually completed, with mesh/convergence evidence

Repository: `[INSERT REPOSITORY URL]` — not fabricated. Creation timestamps and commit history must be genuine. No embedded videos; external links only to real videos.

Slide 13 — Safety and Hazard Mitigation (FMEA)

Full FMEA table in Section 9. No formal ISO/IEEE/IP65 compliance is claimed without test evidence — any IP-rating language used is described as "IP-rated-style sealing approach," not a certified rating.

Slide 14 — Fabrication Roadmap

Full roadmap in Section 10. Stage 3 objective: a safe, working prototype ready for supervised functional field testing — not a promise of final performance. Stage 3 deadline: `[INSERT STAGE 3 DEADLINE]`.

Slide 15 — Technical Challenges and Risks

Full list in Section 11 (shared with Slide 16 — risks and their mitigations are presented together to avoid repeating the same list twice across two slides).

Slide 16 — Risk Mitigation and Backup Solutions

See Section 11 — each risk is paired directly with its backup solution in one combined table, used for both Slide 15 and Slide 16 content.

Slide 17 — References and Sources

Full source list in Section 13.

4. Complete End-to-End Workflow

Covers both the physical process and the electronics/control process, start to finish.

```
Operator checks floats, frame, basket, motors, wiring, enclosure
Operator checks battery, fuse, physical cutoff
Robot placed in controlled test area -> flotation checked
Electronics power on -> SAFE STOPPED STATE (motors off by default)
ESP32-CAM and Arduino Uno establish serial link
Phone connects to ESP32-CAM WiFi -> control page + camera feed checked
Operator issues movement command (F / B / L / R / S)
Arduino validates command
Arduino drives motor-driver logic pins
Left + right motors -> forward / reverse / turn
V-intake guides debris toward basket
Retention mechanism (baffle + lip) reduces escape during turns
Operator monitors, stops manually if jam or instability seen
IF comms timeout (>1s no signal) -> motors auto-stop  [AUTOMATED]
Operator returns robot via manual control
Motors stopped + physical cutoff used BEFORE handling the mechanism
Basket removed, waste sorted, basket cleaned
Operator logs collection efficiency, retention, runtime, failures
Robot inspected, prepared for next trial
```

Branches (automated response vs. operator action, kept explicit):

Condition	Response	Type
Normal operation	Steps 7-13 loop	-
Control/comms loss	Motors stop (firmware timeout, Section 5)	AUTOMATED
Suspected leak	Operator aborts run, returns robot manually	OPERATOR ACTION - no leak sensor in V1
Basket blockage	Operator observes visually, stops and clears	OPERATOR ACTION - no jam sensor in V1

Condition	Response	Type
Abnormal tilt	Operator observes visually and responds	OPERATOR ACTION - tilt is NOT automatically detected unless an MPU6050 is actually installed and wired
Battery/driver problem	Operator notices reduced thrust or heat, stops	OPERATOR ACTION - no battery-voltage monitoring circuit in V1 baseline BOM

This document does not portray tilt detection or battery monitoring as implemented unless the corresponding sensor is actually wired and the firmware actually reads it - both are optional upgrades, not baseline V1.

5. Final Hardware and Software Architecture

5.1 System block diagram (text form - convert to native CAD/schematic before submission; label as "Concept - subject to prototype validation")

```
MECHANICAL                          ACTUATION & POWER
-----------                          -----------------
Twin floats                         Battery (2S Li-ion)
Frame                                  fuse (near battery)
V-shaped intake guides                 physical cutoff switch
Optional PETG baffle insert            voltage regulation
Passive retention mechanism
Removable mesh basket               Motor driver (L298N-class)
Removable hatch                        Motor L        Motor R
                                       (reversible)   (reversible)

ELECTRONICS                         FIRMWARE (Arduino Uno)
------------                         ----------------------
Phone (control page)                Power-on -> SAFE state
   WiFi                             Receive command (serial)
ESP32-CAM (web UI + camera)         Validate command
   Serial (TX/RX, shared GND)       Forward/reverse/turn logic
Arduino Uno (motor logic)           Drive motor-driver pins
   digital pins                     Comms timeout -> auto-stop
Motor driver (above)                Emergency stop (physical)
                                     Operator recovery (manual)
```

No sensor beyond the camera is drawn - turbidity, TDS, temperature, GPS and a wildlife-repellent buzzer remain explicitly excluded from V1 (reasoned through earlier in this project and restated in Section 9).

5.2 Why ESP32-CAM and Arduino Uno are split this way

The ESP32-CAM has very limited free GPIO once its camera and WiFi stack are active - most pins are committed to the camera interface and, if used, the SD card. Driving motors directly from it risks pin conflicts. The validated split: ESP32-CAM handles only camera streaming and the phone control webpage; Arduino Uno handles only motor logic, connected by a simple serial link (single-character commands F/B/L/R/S) with a shared ground. This mirrors the architecture used in a published paper combining an ESP32 and Arduino Uno for a similar boat (Bangladesh solar trash-collector paper, Section 13 Sources).

5.3 Firmware state machine (pseudocode)

```
states: IDLE, FORWARD, BACKWARD, TURN_L, TURN_R, STOP, FAILSAFE

loop:
  if serial_available():
     cmd = read_serial()
     last_cmd_time = now()
     switch(cmd):
       'F' -> state = FORWARD
       'B' -> state = BACKWARD
       'L' -> state = TURN_L
       'R' -> state = TURN_R
       'S' -> state = STOP
  if (now() - last_cmd_time > 1000ms):
     state = FAILSAFE   // no signal = stop motors
  apply_motor_state(state)
```

5.4 Propulsion control logic

Command	Left motor	Right motor
Forward	Forward	Forward
Reverse	Reverse	Reverse

Command	Left motor	Right motor
Turn left	Lower thrust	Higher thrust
Turn right	Higher thrust	Lower thrust
Stop	Off	Off

Propulsion hardware note: baseline is two rear propellers, each on its own motor (assumption, pending bench thrust test). A small paddle-wheel-per-float alternative is an acceptable substitute for aquatic-life-safety reasons (no exposed high-speed underwater propeller, lower entanglement risk) - it is not claimed as project novelty, since paddle-wheel propulsion already appears in a published "Smart Trash Collection Boat" paper (Section 13). Ordinary wheels or a ball-type thruster are not used.

6. Final Mechanical Dimensions and Preliminary Calculations

6.1 Dimensions (subject to CAD verification and physical build)

Part	Dimension	Status
Overall footprint	600 x 450 mm	Assumption
Each float	500 x 95 x 180 mm, sealed	Assumption - actual container/pipe geometry may vary slightly
Basket	300 x 240 x 70 mm, removable mesh	Assumption
Intake opening (between floats)	250-300 mm	Assumption, must equal float inner gap
Entry lip depth below waterline	15-25 mm, tapered (shallower at outer edges for high-floating items like thermocol, deeper at centre for low-floating items like part-filled bottles)	Assumption - needs testing against actual target debris
Mesh gap	3-5 mm, to be validated against target debris sizes	Assumption

6.2 Buoyancy, draft and freeboard calculation

Step 1 - Dry mass estimate (assumption, to be confirmed by weighing the built prototype):

Item	Est. mass
2x PVC floats (shell)	0.6 kg
Frame/mounts	0.5 kg
2x BO motors + propellers	0.15 kg
Battery pack	0.25 kg
Arduino + ESP32-CAM + driver + wiring	0.3 kg
Basket + mesh frame	0.3 kg
Waterproof box + hardware	0.3 kg
Dry mass total	~2.4 kg
+ target waste payload	1.0-1.5 kg
Operating mass	~3.5-3.9 kg

Step 2 - Displacement needed: Buoyant force = weight of displaced water. At 3.9 kg operating mass, 3.9 litres must be displaced (freshwater approx. 1000 kg/m3, so 1 kg needs 1 L).

Step 3 - Float capacity: Each float, fully submerged = 500 x 95 x 180 mm = 8.55 L; both floats = 17.1 L. (Verified - geometric calculation.)

Step 4 - Draft at operating mass: combined float footprint area = 2 x (0.5 x 0.095) = 0.095 sq m. Draft = 0.0039 cu m / 0.095 sq m = approx. 41 mm, roughly 23% of the 180 mm float height submerged, leaving approx. 139 mm freeboard. (Assumption - calculated from Step 1-3; confirm with loaded flotation test.)

Step 5 - Safety margin: even at double the estimated mass (approx. 8 kg), draft only reaches approx. 90 mm, roughly 50% of float height. The floats carry meaningful reserve buoyancy; this margin is deliberate and should not be reduced to save material cost.

6.3 CAD specification (condensed from the project's working CAD brief)

Twin-float catamaran, 600x450 mm footprint. Two sealed PVC floats, 500x95x180 mm, 250-300 mm gap matching the intake/basket width. Waterline marked at approx. 41 mm draft / approx. 139 mm freeboard at operating mass. V-funnel tapering from the full float gap to a 150-180 mm basket mouth. Tapered entry lip (15-25 mm). Static retention baffle (Section 8) immediately behind the lip, no hinge. Basket near-horizontal (not a steep ramp), positioned directly under a full-width hatch, with visible drain gaps in the mesh floor. Electronics enclosure mounted low and centred between the floats, sealed by a physical bulkhead from the wet basket zone. Two motors, one per float, explicit twin-motor differential steering. Label every CAD render "Concept CAD - subject to prototype validation."

6.4 Structural load notes (no FEA performed - explicitly stated)

Critical load-bearing areas: basket support rails (must carry approx. 1.5 kg eccentric load without visible deflection), float-to-frame mounting points, motor mounts. FEA workflow if later performed: static load case on basket rails at 1.5 kg eccentric load, frame mount points fixed as boundary condition, mesh-convergence check before trusting results. Until then, the physical equivalent is a loaded-basket deflection test (hang 1.5 kg off-centre in the assembled basket, measure sag) - this substitutes for FEA in V1 and is the only load evidence this document claims.

7. Itemized Budget and Procurement Status

New-purchase budget target: Rs 8,000-10,000 (floor version achievable near Rs 5,000). Arduino Uno and ESP32-CAM already owned, excluded from totals below. Prices are current supplier-listed estimates from Indian electronics/robotics suppliers (IndiaMART-type listings, probots.co.in) found during research - not guaranteed quotes. Confirm before purchase.

Item	Spec	Qty	Est. unit price (Rs)	Est. total (Rs)	Status	Source note
BO geared motor, 100 RPM, 12V	dual-shaft, plastic gear	2	100	200	To purchase	BO series dual-shaft 100 RPM motor, listed on probots.co.in

Item	Spec	Qty	Est. unit price (Rs)	Est. total (Rs)	Status	Source note
Propellers (small boat-type) or DIY from bottle caps	-	2	50	100	To purchase	estimate
L298N dual motor driver module	-	1	130	130	To purchase	L298N module listed around Rs 100-140/piece on IndiaMART
Battery: 2S 18650 Li-ion (7.4V, approx 2200mAh) + holder	-	1	350	350	To purchase	estimate, verify local seller
TP4056 charging module	-	1	40	40	To purchase	estimate
In-line fuse + holder (3A)	-	1	40	40	To purchase - safety	estimate
Toggle switch + emergency push-stop button	-	2	50	100	To purchase - safety	estimate
PVC pipe (110mm dia) + 4 end caps + solvent cement	for 2 floats	1 set	500	500	To purchase	estimate
Plywood/acrylic sheet for frame	3-5mm	1 sheet	350	350	To purchase	estimate
PETG or acrylic sheet for baffle insert	thin, 1-2mm	1 sheet	400	400	To purchase	estimate - PETG preferred, more impact tolerant
L-brackets, nuts, bolts, fasteners	-	assorted	200	200	To purchase	estimate
Stainless kitchen mesh / net bag for basket	3-5mm gap	1	150	150	To purchase	estimate
Small waterproof box (lunch-box type) + silicone sealant	-	1	250	250	To purchase - electronics protection	estimate
Cable glands	-	4	20	80	To purchase	estimate

Item	Spec	Qty	Est. unit price (Rs)	Est. total (Rs)	Status	Source note
Jumper wires, perfboard, heat-shrink, connectors	-	-	200	200	To purchase	estimate
Retrieval tether (cord)	-	1	50	50	To purchase	estimate
Subtotal				approx Rs 3,140		
Contingency / shipping (15%)				approx Rs 470		
Total new-purchase estimate				approx Rs 3,600-4,600		Leaves buffer under the Rs 8,000-10,000 target

If the final design (after prototyping the baffle, Section 8) exceeds this estimate, the specific cost reduction is: drop the PETG baffle insert and use the plain V-intake plus a fixed foil-lip only (removes Rs 400 and the baffle fabrication step) - never reduce the fuse, cutoff switch, or waterproofing budget to save cost.

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8. Novelty and Prior-Art Assessment

8.1 What already exists (do not claim as novel)

- A floating waste collector - exists widely (Seabin, WasteShark, multiple student projects)
- Twin floats / catamaran hull - used by KSCST's river waste collector, EcoBoat, WasteShark
- Differential steering with two motors - standard RC-boat technique
- A camera on a collection boat - used in the Bangladesh solar trash-collector paper and others
- A removable basket - standard across nearly all reviewed projects
- A one-way flap / retention gate - patented decades ago in swimming-pool skimmer designs (foil-lip "double trap")

- A Tesla-valve-inspired retention geometry for floating debris - already published (EcoBoat, 2024/25 research paper, arxiv) at a 15kg+, 1700x900x600mm aluminium-framed scale
- A mobile-phone controller for a trash boat - used in the Nepal diploma project and the Bangladesh paper via Bluetooth/WiFi

8.2 The actual engineering gap identified

Nearly every reviewed Indian student trash-collection boat (KSCST river waste collector, J.P Institute river cleaning boat, Nepal diploma Trash Collection Bot, the 2026-published "Smart Trash Collection Boat" paper, the Bangladesh solar-powered trash robot paper) uses a powered conveyor belt requiring 2 or more dedicated motors on top of propulsion motors. None of the reviewed sources report a measured collection-efficiency percentage - only payload capacity and runtime. None of the reviewed sources (including WasteShark's public material) document how well a passive retention mechanism performs specifically during turning or reversing.

8.3 Defensible project contribution

"VARUNA AquaHalo investigates a low-cost, conveyor-free surface-waste collector using a modular V-intake and a passive, no-moving-parts retention mechanism - adapted from a published macro-scale Tesla-valve-inspired debris-retention concept (EcoBoat) and miniaturized to a sub-Rs 10,000, two-motor, student-buildable scale - with intake and retention performance evaluated through repeatable collection, retention, stability and runtime tests, rather than assumed."

This statement remains accurate after the research conducted for this project (Section 13 Sources) and is used as the core innovation statement on Slide 1.

8.4 Demonstrated features vs. future work (kept clearly separate)

Demonstrated in V1 (to be built and tested)	Future work (explicitly NOT V1)
Conveyor-free V-intake + retention baffle	GPS, autonomous navigation
Removable mesh basket	AI/computer-vision waste recognition
Phone-based manual control via ESP32-CAM	UV-C or any water-treatment module
Measured collection efficiency and retention rate	Turbidity/TDS/temperature sensing

Demonstrated in V1 (to be built and tested)	Future work (explicitly NOT V1)
Twin-motor differential steering	Solar charging
Basic comms-loss failsafe (motor stop)	Multi-unit coordinated operation

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9. Safety and FMEA

No formal ISO, IEEE or IP65 compliance is claimed without test evidence. Any sealing description used in this project is phrased as "an IP-rated-style sealing approach," never a certified rating.

Failure mode	Cause	Effect	Severity	Likelihood	Detection	Mitigation	Residual risk
Float leakage	Poor PVC solvent joint, crack from handling	Reduced buoyancy, possible sinking	High	Medium	Visual waterline check, soak test before each session	Pressure/soak-test each float separately before assembly; inspect before every launch	Low after testing, re-check periodically
Electronics water ingress	Enclosure seal failure, cable gland gap	Short circuit, loss of control	High	Medium	Visual inspection, unexpected behaviour during test	Sealed box, silicone sealant, cable glands with strain relief	Low-medium, monitor over repeated use
Motor driver overheating	Stall current exceeds driver rating, jam	Driver failure, possible fire risk at small scale	Medium	Low-medium	Heat felt on casing, erratic motor behaviour	Match driver to measured stall current (not just running current); add fuse	Low with correct driver selection
Battery short circuit	Damaged cell, wiring fault	Fire/burn risk, battery damage	High	Low	Unusual heat, smell	Fuse near battery, physical cutoff, careful wiring, TP4056 protected charging	Low with fuse in place
Lost phone/WiFi connection	Range exceeded, interference	Boat continues on last command indefinitely	Medium	Medium	No response to new commands	Firmware comms-loss timeout (1s) auto-stops motors	Low - automated mitigation in firmware

Failure mode	Cause	Effect	Severity	Likelihood	Detection	Mitigation	Residual risk
Uncommanded propulsion	Firmware bug, noise on serial line	Unexpected movement	Medium	Low	Visual observation	Safe power-on state (motors off by default), validated command parsing	Low
Basket jamming	Debris wedged in retention baffle or entry lip	Reduced collection, possible drag increase	Medium	Medium	Visual observation, reduced boat speed	Slot-in adjustable baffle stages (Section 8), designed for easy clearing	Medium until baffle geometry is tuned by testing
Excessive drag	Baffle geometry too restrictive, basket fouling	Reduced speed/control, higher current draw	Low-medium	Medium	Measured speed drop, current draw increase	Bucket-test baffle before final install; fall back to plain V-intake if drag is excessive	Low with fallback plan
Uneven loading / instability	Waste collected unevenly in basket, battery placement	Listing, reduced freeboard on one side	Medium	Medium	Visual tilt observation	Low, central battery placement; centred basket; loaded flotation test	Medium until confirmed by test
Propeller entanglement	Fishing line, long weeds	Motor stall, possible motor damage	Medium	Medium	Reduced thrust, motor strain	Paddle-wheel alternative considered specifically for this risk (Section 5.4); propeller guard if propellers are used	Medium - best addressed by paddle-wheel choice
Recovery failure	Boat drifts out of control range or control lost in open water	Boat stranded, retrieval difficulty	Medium	Low-medium	Loss of control signal	Retrieval tether during early water tests; testing confined to supervised, contained area	Low with tether and supervision

Safety provisions included in the baseline design: physical emergency cutoff, battery fuse near the battery, motor driver matched to stall current, communication-loss timeout, motors default-stopped on power-on, waterproof electronics enclosure, cable glands with strain relief, motor/propeller guard where practical, retrieval tether during early tests, and testing confined to a controlled area away from swimmers, wildlife and other boat traffic.

10. Fabrication Roadmap

Stage 3 objective: a safe, working prototype ready for supervised functional field testing - not a promise of final performance. Stage 3 deadline: [INSERT STAGE 3 DEADLINE].

Week	Tasks	Deliverable	Budget spent	Success criteria
Week 1	Requirements confirmation, CAD modelling, component selection, budget confirmation, order long-lead items	CAD model v1, finalized BOM, components ordered	~Rs 1,000	CAD dimensions match Section 6; BOM total within budget
Week 2	Frame, floats, basket and intake fabrication; float leak testing	Assembled hull with floats, leak-tested	~Rs 1,800	Floats pass soak test; frame assembled square and rigid
Week 3	Electronics mounting, motor mounting, wiring, firmware integration	Wired, powered prototype; firmware uploaded	~Rs 1,800	Motors respond to phone commands; failsafe verified on bench
Week 4	Dry testing, loaded buoyancy test, controlled collection trials, documentation	Tested prototype, test logs	Remaining contingency	Collection efficiency and retention rate measured and logged

11. Testing Protocol and Impact Metrics

All results in this document are 'Testing pending' until measured. No simulated numbers substitute for real results.

11.1 Defined metrics

- Collection efficiency (%) = (items collected / items released) x 100
- Retention rate (%) = (items retained after a defined turning test / items initially collected) x 100
- Collection rate = items collected / test duration

- Runtime = time from start of operating test to the chosen safe stopping condition
- Stability = loaded waterline, freeboard, visible tilt, water ingress at empty / half-loaded / fully loaded / unevenly loaded states
- Control safety = motors stop when commands are deliberately interrupted

11.2 Retention mechanism comparison (core experimental evidence)

1. Baseline: plain V-intake, no retention feature - release 10 items, perform 3x full 360-degree turns, count escapes
2. Mechanism test: same debris, same turns, with retention baffle and foil lip installed
3. Report retention rate for both, same debris mix, same duration, minimum 3 repetitions

If the mechanism does not outperform the baseline, that result is reported honestly rather than omitted.

11.3 Results template (fill with measured data only)

Test	Condition	Repetitions	Measured result	Status
Collection efficiency	Plain V-intake	3	TBD	Testing pending
Collection efficiency	V-intake + baffle	3	TBD	Testing pending
Retention after 360 deg turn	Plain V-intake	3	TBD	Testing pending
Retention after 360 deg turn	V-intake + baffle	3	TBD	Testing pending
Runtime	Full charge to shutdown	1-2	TBD	Testing pending
Comms-loss failsafe	Signal deliberately cut	3	TBD	Testing pending

12. Technical Challenges and Backup Solutions

Risk / trade-off	Preventive design change	Detection test	Backup solution	Cost/complexity impact
Motor thrust vs basket drag	Baffle gaps sized against largest target debris (Section 6)	Bench thrust test with baffle installed	Revert to plain V-intake with fixed foil-lip only	Low - removes PETG baffle cost
Float size vs stability and weight	Floats sized with deliberate reserve buoyancy (Section 6.2)	Loaded flotation test	Do not shrink floats to save cost	None - already accounted for
Battery runtime vs motor current	2S Li-ion sized for estimated draw	Timed discharge test under load	Add a second parallel cell if runtime is too short	+Rs 300-400
Basket immersion vs collection performance	Entry lip depth tapered, not fixed	Water test with real target debris	Adjust lip depth per debris type observed	Low - rebuild lip only
Retention mechanism vs jamming	Slot-in, adjustable baffle stages, not glued	Dry flap/baffle test before water test	Simplify to foil-lip only if baffle jams	Low
Phone connectivity vs control reliability	Comms-loss failsafe in firmware	Range walk test, deliberate signal cut	Manual recovery procedure regardless of connectivity state	None - already designed in
Waterproofing vs maintenance access	Sealed box with cable glands, not glued shut	Visual inspection after each test	Re-seal and retest before next session	Low

Risk / trade-off	Preventive design change	Detection test	Backup solution	Cost/complexity impact
PETG baffle geometry vs debris blockage	Bucket/tub test before final CAD	10-minute bucket test with real debris	Adjust fin angle/gap or drop to foil-lip only	Low
Rs 8,000-10,000 budget vs component quality	BOM built with contingency (Section 7)	Running total check at each purchase	Remove optional upgrades before reducing safety provisions	None if followed in order

If the budget is exceeded, optional features are removed before any safety provision (fuse, cutoff switch, waterproofing) is reduced. If the controller disconnects, propulsion stops automatically (already solved in firmware, Section 5.3).

13. Sources

Source	Publisher/repository	What it informed
EcoBoat: Design and Experimental Validation of an Autonomous Body-Board Boat For Cleaning Water Bodies	arXiv	Tesla-valve-inspired retention basket concept - informed the retention mechanism, scaled down from its 15kg/1700mm platform
Smart Trash Collection Boat	ICADMSES 2026 proceedings	Confirms conveyor + paddle-wheel pattern in comparable student projects; figures used as prior-art comparison
Design of River Waste Water Collector Using Arduino	KSCST project synopsis	Catamaran-hull, conveyor-based prior art comparison

Source	Publisher/repository	What it informed
Trash Collection Bot	Kathmandu University KURC	Conveyor plus Bluetooth control prior-art comparison; styrofoam-buoyancy risk noted
Solar Powered Water Trash Collector Robot	Independent University, Bangladesh repository	ESP32 plus Arduino Uno plus L298N architecture pattern, informed board-split decision
Swimming pool skimmer, US Patent 4820411	US Patent and Trademark Office	Fixed foil-lip double-trap retention principle
Seabin Project coverage	Trade press	Confirms Seabin is fixed, pump-suction, dock-mounted - informed decision against suction design
Autonomous Water Cleaner lagoon skimmer	Florida Atlantic University senior design	Additional catamaran plus conveyor prior-art comparison
A State-of-the-Art Compact Surface Drifter Reveals Pathways of Floating Marine Litter in the German Bight	Frontiers in Marine Science, 2019	Wind-driven surface current behaviour, informed deployment-strategy recommendation
9.7 Langmuir Circulation	Introduction to Oceanography open textbook	Langmuir circulation and debris-streak accumulation, informed deployment strategy

Component prices in Section 7 are sourced from IndiaMART-type listings and probots.co.in, identified as supplier-listed estimates requiring confirmation, not verified final retail prices.

14. Final Pre-Submission Checklist

- [ ] Team ID and official track confirmed and inserted
- [ ] CAD model matches Section 6 dimensions, labelled Concept CAD until fabricated

- [ ] BOM total confirmed against actual supplier prices before purchase
- [ ] Motor driver selected against measured stall current, not assumed
- [ ] Baffle geometry bucket-tested before committing to final CAD
- [ ] Loaded flotation test performed, Section 6.2 updated with measured values
- [ ] Comms-loss failsafe verified on bench before first water test
- [ ] Retrieval tether used for all early water tests
- [ ] Collection efficiency and retention-rate tests run and logged
- [ ] No fabricated CAD screenshots, FEA results, photographs, or repository evidence included
- [ ] Repository URL inserted once a real repository exists
- [ ] Stage 3 deadline inserted once supplied
- [ ] Every claim checked against this document's status labels before submission